Predicting Five-Year Charge-Off Risk in SBA 7(a) Loans

1. Project Title and Disciplinary Connection

Predicting Five-Year Charge-Off Risk in SBA 7(a) Loans

This project connects data analytics with finance and credit-risk analysis. I am interested in how historical lending data can be used to understand credit performance and support lending decisions. The project applies classification methods to small-business loans guaranteed through the U.S. Small Business Administration (SBA) 7(a) program.

2. Research Question

Using information available around loan approval, how well can we predict whether an SBA 7(a) loan will be charged off within five years of approval?

The main goal of the project is **predictive**. Each observation represents one SBA 7(a) loan. The positive class is a loan that experiences a documented charge-off within five years after its approval date.

I chose a fixed five-year outcome because loans approved in different years need the same amount of time to experience a charge-off. Since the current SBA data are reported through June 30, 2026, the modeling sample will only include loans approved on or before June 30, 2021.

3. Motivation and Stakeholders

Credit risk is an important part of commercial lending because lenders need to identify loans that may have a higher chance of future loss. I want to study whether characteristics already known around approval contain useful information about later loan performance.

Potential users of this analysis include credit analysts, lenders, portfolio-risk teams, and SBA program stakeholders. The model is not intended to replace underwriting

or automatically decide whether a borrower should receive credit. Instead, I will treat it as a **risk-screening model** that shows how much predictive information is available in the public SBA data.

A false negative would mean that the model does not flag a loan as high risk even though it later charges off. A false positive would mean that the model flags a loan as high risk even though it does not charge off within five years. Both types of mistakes matter because one may result in missed credit risk while the other may lead to unnecessary additional review.

4. Data and Access Status

I will use the **U.S. Small Business Administration 7(a) and 504 FOIA dataset**, specifically the 7(a) loan files. The SBA currently provides separate loan-level files for FY2010–FY2019 and FY2020–Present, along with an official data dictionary. The files are updated quarterly; the current release is reported through June 30, 2026.

One row represents one SBA 7(a) loan. Potential predictors include gross approval amount, SBA-guaranteed amount, loan term, initial interest rate, NAICS industry, project state, business age and type, processing method, collateral indicator, revolving status, and franchise information.

Outcome variables such as LoanStatus, ChargeOffDate, and GrossChargeOffAmount will only be used to construct or verify the response variable. They will not be used as predictors because doing so would introduce information that was not available at approval.

I have confirmed that the official files and data dictionary are publicly available and have prepared the code for downloading, cleaning, and analyzing the files. Because the files are large, the analysis will use local or Google Drive copies rather than repeatedly downloading them.

5. Planned Analysis

I will first examine data types, duplicates, missing values, distributions, and unusual observations. Exploratory analysis will look at how five-year charge-off rates vary by variables such as loan amount, term, interest rate, industry, approval year, and SBA guarantee percentage.

The data will be separated chronologically rather than randomly:

- **2010–2017:** training data
- **2018–2019:** validation and model selection
- **2020–June 2021:** locked test data

This structure allows the model to learn from older loans and then be evaluated on later loans, which is closer to how a credit-risk model would be used in practice.

Numeric missing values will be imputed using training-set medians and standardized when appropriate. Categorical variables will be imputed and one-hot encoded. These steps will be included inside a Scikit-learn preprocessing pipeline so information from validation or test data does not enter the training process.

I plan to compare a **Dummy Classifier, Logistic Regression, Decision Tree, and Random Forest**. The Dummy Classifier will provide a simple baseline, while Logistic Regression provides an interpretable credit-risk benchmark.

Because charge-offs are expected to be less common than non-charge-offs, **PR-AUC (Average Precision)** will be the main metric for comparing models. I will also report ROC-AUC, precision, recall, F1 score, and a confusion matrix. The classification threshold will be selected using validation data rather than the test set. I will also examine the training-validation performance gap, a learning curve, and examples of false positives and false negatives to better understand model errors.

6. Feasibility Assessment and Backup Plan

The project is manageable because it has one specific prediction target, publicly available data, and a limited set of models.

The largest risk is constructing a reliable five-year charge-off outcome. Some charged-off loans may have incomplete dates, and loans approved after June 2021 have not yet had a full five-year observation period. I will therefore exclude recent loans from the modeling sample and exclude cases where the timing of a charge-off cannot be determined reliably.

Another limitation is that the SBA public data do not contain all of the borrower financial information that a bank would normally use in underwriting, such as cash

flow, leverage, debt-service coverage, detailed financial statements, or credit history. Therefore, the model should be interpreted as a simplified credit-risk model based on publicly available loan characteristics.

If missing charge-off dates make the five-year outcome unreliable for too many observations, my backup plan is to narrow the sample further to older loan cohorts with clearer performance information and redefine the outcome using documented loan status. If necessary, I can also reduce the number of predictors or models without changing the main credit-risk question.

7. Timeline

September 17–21: Download and verify the SBA files, finalize the five-year target, clean the data, and document missing-data and data-quality issues.

September 22–28: Complete exploratory analysis, feature engineering, visualizations, and the first instructor meeting.

September 29–October 4: Build baseline and classification models, compare validation performance, choose a threshold, perform error analysis, and complete the second instructor meeting.

October 5–8: Finalize the model results and prepare the in-class presentation.

October 9–15: Complete the written report, limitations and interpretation, clean the GitHub repository, test reproduction steps, and submit the final materials.

8. References and Links

U.S. Small Business Administration. **7(a) & 504 FOIA Loan Data.** SBA Open Data. The dataset contains loan-level 7(a) records and an accompanying data dictionary and is updated quarterly.

Géron, Aurélien. *Hands-On Machine Learning with Scikit-Learn and PyTorch.* Chapters 2–4 and companion notebooks.

GitHub repository:
https://github.com/chauphamcreditproduct/sba-credit-risk-mini-project